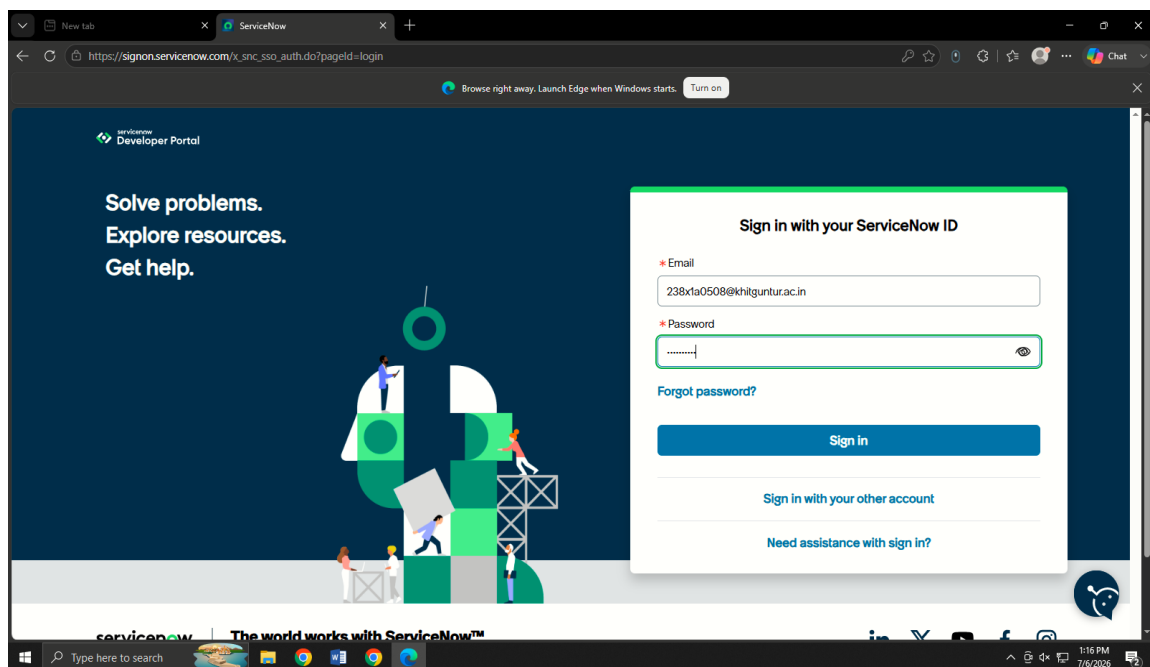


Optimizing User, Group, and Role Management with Access Control and Workflows

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Project Name	Optimizing User, Group, and Role Management with Access Control and Workflows
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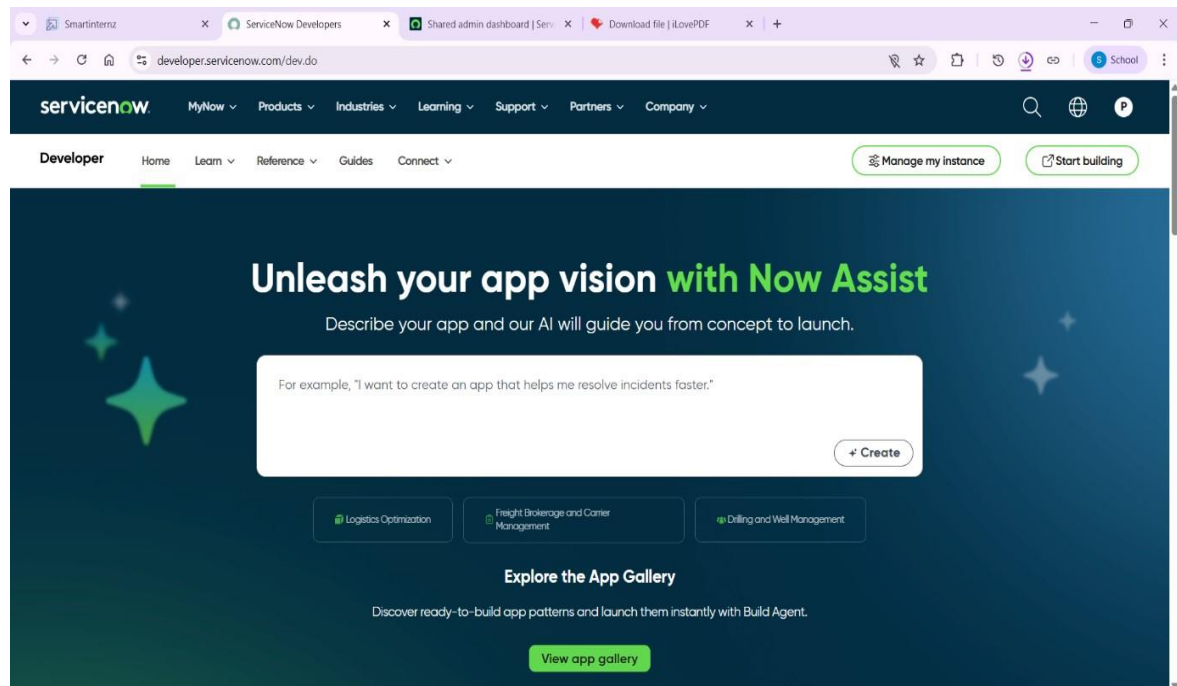
Step 1: Setting Up ServiceNow Instance

- Go to the ServiceNow Developer Site. (<https://developer.servicenow.com/>)
 - Sign In to the Developer Account.



- Request a Personal Developer Instance (PDI)
 1. Click on the profile icon at the top-right corner.
 2. Select "Manage Instance" or go directly to the link.
 3. Click "Request Instance".
 4. Choose the latest version (default selection is recommended).

- Access your Instance You'll get an instance URL (https://dev225052.service-now.com) Click "Open Instance" to launch it.
- Your admin credentials (username & password) will be shown-copy and save them securely.
- Log in to Your Instance.



Usage:

- Use the Application Navigator (left sidebar) to explore modules like:
Incident, Problem, Change Management.
Service CatLog.
Configuration Management (CMDB).
Workflow Editor / Flow Designer.
- Use App Engine Studio or ServiceNow Studio to build or customize apps.

Step 2: Create Users

A user in ServiceNow is an individual record (stored in the sys_user table) representing a person who can log in to the instance to access data, submit requests, or manage tasks. These accounts define identity, roles, and preferences, allowing personalized access to the platform.

- Open service now.
- Click on All >> search for **users**.

- Select Users under system security.
- Click on **new** and enter the following details of **User1 of User2:**

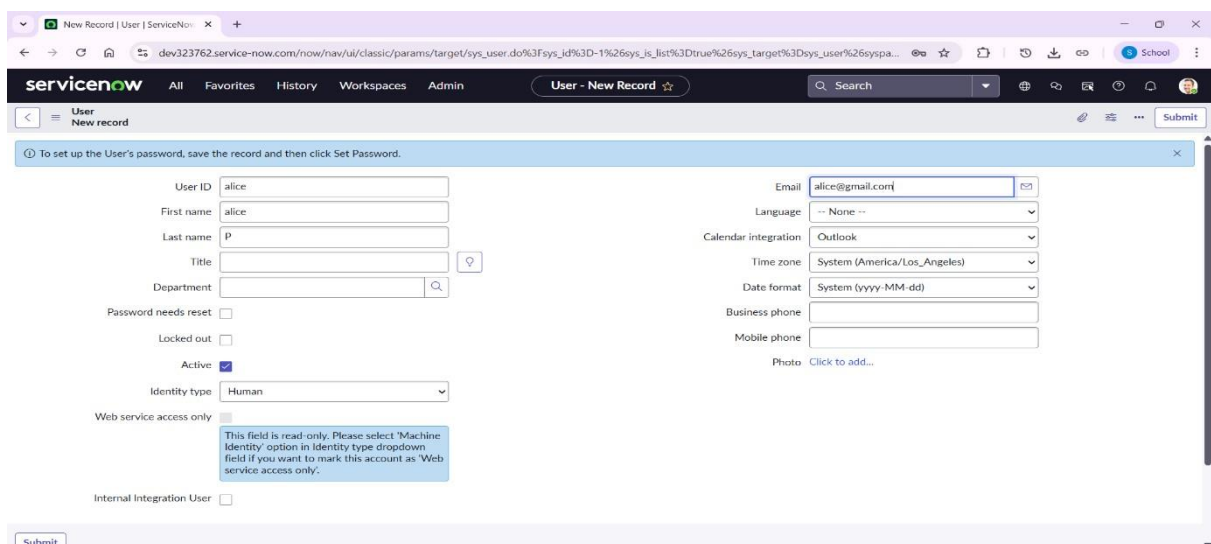
User1:

- ❖ **User ID:** alice
- ❖ **First Name:** alice
- ❖ **Last Name:** P
- ❖ **Email:** alice@gmail.com

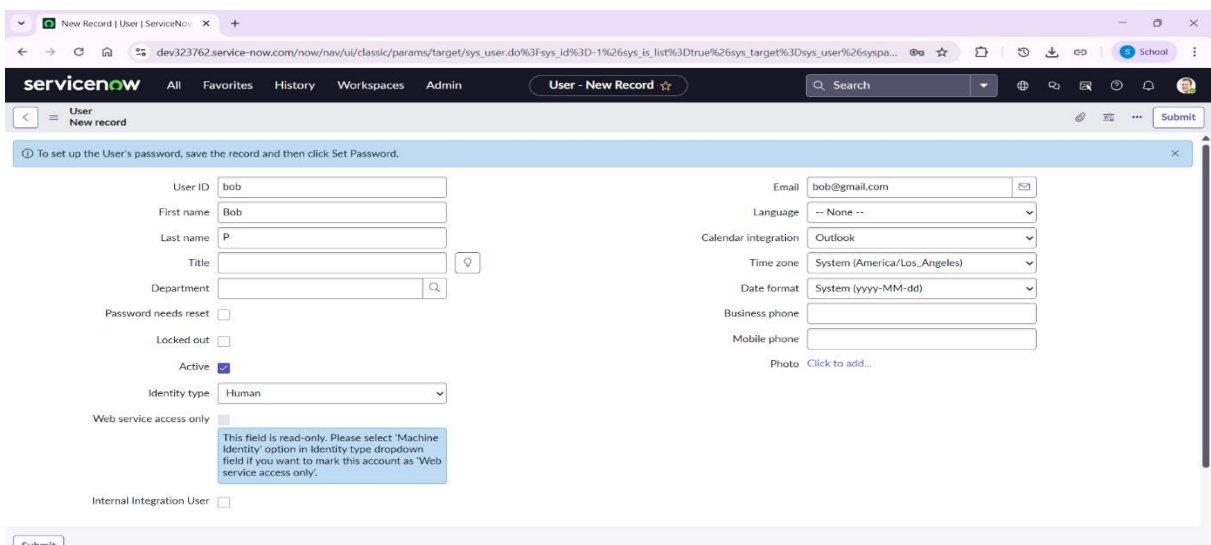
User2:

- ❖ **User ID:** bob
- ❖ **First Name:** Bob
- ❖ **Last Name:** P ❖ **Email:** bob@gmail.com

- After entering all details click **Submit**.



The screenshot shows the 'User - New Record' form in ServiceNow. The form is titled 'User - New Record' and has a 'Submit' button at the top right. A blue banner at the top states: 'To set up the User's password, save the record and then click Set Password.' The form fields are organized into two columns. The left column contains: 'User ID' (alice), 'First name' (alice), 'Last name' (P), 'Title' (empty), 'Department' (empty), 'Password needs reset' (checkbox), 'Locked out' (checkbox), 'Active' (checkbox, checked), 'Identity type' (Human), 'Web service access only' (checkbox), and 'Internal Integration User' (checkbox). The right column contains: 'Email' (alice@gmail.com), 'Language' (None), 'Calendar integration' (Outlook), 'Time zone' (System (America/Los Angeles)), 'Date format' (System (yyyy-MM-dd)), 'Business phone' (empty), 'Mobile phone' (empty), and 'Photo' (Click to add...). A blue tooltip is visible over the 'Web service access only' checkbox, stating: 'This field is read-only. Please select 'Machine Identity' option in Identity type dropdown field if you want to mark this account as 'Web service access only'.'



The screenshot shows the 'User - New Record' form in ServiceNow, identical to the one above but with the details for User2 (bob). The form fields are: 'User ID' (bob), 'First name' (Bob), 'Last name' (P), 'Title' (empty), 'Department' (empty), 'Password needs reset' (checkbox), 'Locked out' (checkbox), 'Active' (checkbox, checked), 'Identity type' (Human), 'Web service access only' (checkbox), and 'Internal Integration User' (checkbox). The right column contains: 'Email' (bob@gmail.com), 'Language' (None), 'Calendar integration' (Outlook), 'Time zone' (System (America/Los Angeles)), 'Date format' (System (yyyy-MM-dd)), 'Business phone' (empty), 'Mobile phone' (empty), and 'Photo' (Click to add...). The same blue tooltip is visible over the 'Web service access only' checkbox.

Step 3: Create Groups

A group in ServiceNow is a foundational, (defined in the sys_user_group table) used to organize users with shared purposes, such as team members, for assigning tasks, roles, and permissions. Groups streamline administration by allowing roles (like ITIL) and permissions to be assigned to the group, which are inherited by all members. They are critical for workflow, including incident assignment, approval processes, and email notifications.

- Open service now.
- Click on All >> search for **groups**.
- Select Users under system security.
- Click on **new** and enter the following details.

❖**Name:** Project Team.

- After entering all details click **Submit**.

The screenshot displays the ServiceNow interface for creating a group. The browser address bar shows the URL: dev323762.service-now.com/now/nav/ui/classic/params/target/sys_user_group.do%3Fsys_id%3D3De28c9856477632100aa044c0806d4327%26sysparm_re... The page title is 'Project Team | Group | ServiceNow'. The breadcrumb trail is 'All > Favorites > History > Workspaces > Admin > Group - Project Team'. The form fields are: Name (Project Team), Group email (empty), Manager (empty), Parent (empty), and Description (empty). Below the form are 'Update' and 'Delete' buttons. A tabbed interface shows 'Roles', 'Group Members (2)', and 'Groups'. The 'Roles' tab is active, displaying a table with columns: Created, Role, Granted by, and Inherits. The table is currently empty, showing only a header and a search bar. A 'Created' dropdown menu is visible above the table.

Step 4: Create Roles

In ServiceNow, a role is a collection of permissions stored in the sys_user_role table that determines what a user can access, view, or modify within the platform. Roles act as a security mechanism (RBAC) to ensure users have the necessary permissions for their job functions.

- Open service now.
- Click on All >> search for **roles**.
- Select Users under system security.
- Click on **new** and enter the following details of **Role1** of **Role2**:

Role1:

❖ **Name:** Project Member.

- After entering all details click **Submit**.

The screenshot shows the ServiceNow interface for creating a role named 'Project Member'. The form includes fields for 'Name' (Project Member), 'Application' (Global), and 'Description'. There is an 'Elevated privilege' checkbox. Below the form are 'Update' and 'Delete' buttons. A tabbed interface shows 'Contains Roles', 'Applications with Role', 'Modules with Role', and 'Custom Tables'. The 'Contains Roles' tab is active, showing a search bar and a table with no records displayed.

Role2:

❖ **Name:** Team Member.

- After entering all details click **Submit**.

The screenshot shows the ServiceNow interface for creating a role named 'Team Member'. The form includes fields for 'Name' (Team Member), 'Application' (Global), and 'Description'. There is an 'Elevated privilege' checkbox. Below the form are 'Update' and 'Delete' buttons. A tabbed interface shows 'Contains Roles', 'Applications with Role (1)', 'Modules with Role', and 'Custom Tables'. The 'Applications with Role (1)' tab is active, showing a search bar and a table with no records displayed.

Step 4: Create Table

In ServiceNow, a table is the fundamental database structure for storing data, similar to a spreadsheet or database table, composed of fields (columns) that hold specific data types and records (rows) that represent individual entries, like a user or an incident. Every action in

ServiceNow, from creating an incident to managing users, involves interacting with data stored in these tables, which form the backbone of applications and configurations.

- Open service now.
- Click on All >> search for **tables**.
- Select tables under system definition.
- Click on **new**.
- Enter the following details to create a new table:

Table1:

- ❖ **Label:** project table.
- ❖ **Check the boxes:** Create module & Create mobile module.
- ❖ **Under new menu name:** project table.
- Under table columns give the columns:

Column Label	Type
Project ID	Integer
Project Name	String
Project Manager	String
Start Date	Date
End Date	Date
Status	Choice
Description	String

- After entering all details click **Submit**.

Table - New Record

* Label: Project Table

* Name: u_project_table

Extends table: [Search]

Application: Global

Create module: ☒

Create mobile module: ☒

Add module to menu: -- Create new --

New menu name: Project Table

Columns | Controls | Application Access

Table Columns [for text] [Search]

Dictionary Entries

	Column label	Type	Reference	Max length	Default value	Display
✕ ✎	Project ID	Integer				false
✕ ✎	Project Name	String				false
✕ ✎	Project Manager	String				false
✕ ✎	Start Date	Date				false
✕ ✎	End Date	Date				false
✕ ✎	Status	Choice				false
✕ ✎	Description	String				false

Table2:

- ❖ **Label:** task table 2.
- ❖ **Check the boxes:** Create module & Create mobile module.
- Under table columns give the columns:

Column Label	Type
Task ID	Integer
Task Name	String
Assigned To	String
Due Date	Date
Status	Choice
Comments	String
- After entering all details click **Submit**.

Table - New Record

* Label: Task Table 2

* Name: u_task_table_2

Extends table: [Search]

Application: Global

Create module: ☒

Create mobile module: ☒

Add module to menu: -- Create new --

New menu name: Task Table 2

Columns Controls Application Access

Table Columns for text Search

Dictionary Entries	Column label	Type	Reference	Max length	Default value	Display
☒	Task ID	Integer			false	false
☒	Task Name	String			false	false
☒	Assigned To	String			false	false
☒	Due Date	Date			false	false
☒	Status	Choice			false	false
☒	Comments	String			false	false

Insert a new row...

Step 5: Assign users to groups

Assigning users to groups in ServiceNow is a fundamental administrative action used to organize users for task routing, role-based access control, and approvals. Users are assigned via the sys_user_grmember table (Group Members) by navigating to User Administration > Groups, selecting a group, and adding users in the Members related list.

- Open servicenow.
- Click on All >> search for **groups**.
- Select tables under system definition.
- Select the project team group.
- Under group members.
- Click on **edit** and Select alice p and bob p and **save**.

Group - Project Team

Name: Project Team

Group email: [Search]

Manager: [Search]

Parent: [Search]

Description: [Text Area]

Update Delete

Roles Group Members (2) Groups

User Search

Group = Project Team

User
alice P
Bob P

1 to 2 of 2

Step 6: Assign roles to users

Assigning roles to users in ServiceNow is the process of granting specific permissions or access rights to a user profile, directly or via groups, controlling their ability to view or modify data. This is managed by administrators in the sys_user table, enabling access to applications and modules.

❖ Assign roles to alice user:

- Open servicenow.
- Click on All >> search for user
- Select tables under system definition
- Select the project manager user
- Under project manager
- Click on edit and Select project member and save
- click on edit add u_project_table role and u_task_table role •
- click on save and update the form.

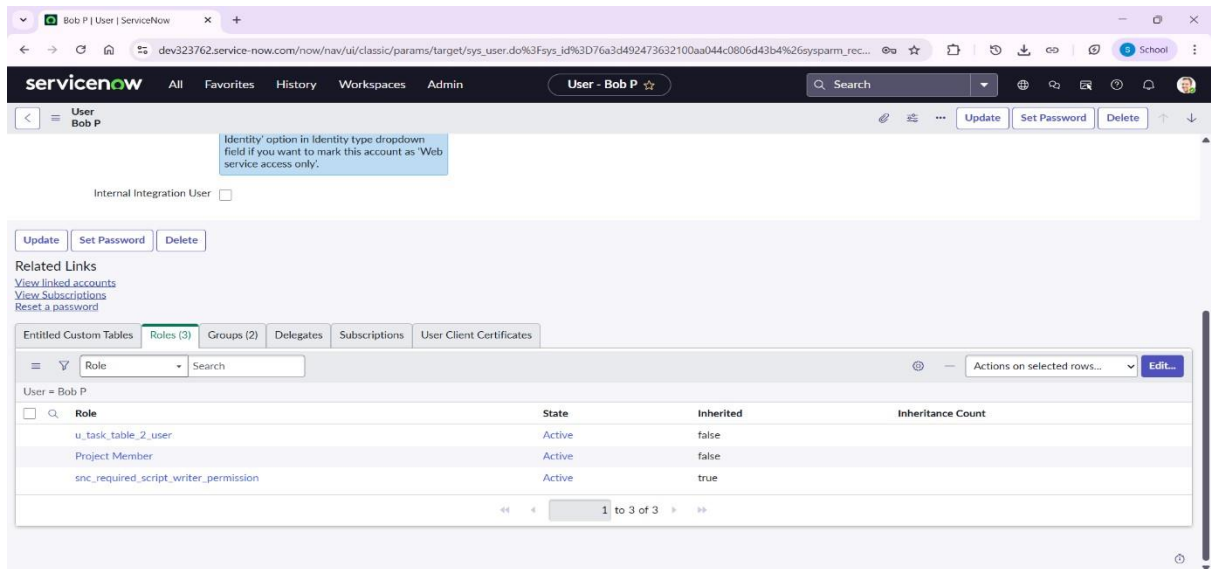
The screenshot shows the ServiceNow user management interface for user 'alice P'. The 'Roles' tab is selected, displaying a table of assigned roles. The roles listed are Project Member, u_project_table_user, snc_required_script_writer_permission, and u_task_table_2_user. The 'snc_required_script_writer_permission' role is highlighted, showing it is inherited (true) and has an inheritance count of 1.

Role	State	Inherited	Inheritance Count
Project Member	Active	false	
u_project_table_user	Active	false	
snc_required_script_writer_permission	Active	true	1
u_task_table_2_user	Active	false	

❖ Assign roles to bob user • Open servicenow.

- Click on All >> search for **user**.
- Select tables under system definition.

- Select the bob p user.
- Under team member.
- Click on **edit**.
- Select team member and give table role and save.
- Click on profile icon Impersonate user to bob.
- We can see the task table2.



Step 7: Application access

Application Access in ServiceNow refers to settings that control how custom applications and their data (specifically tables) are accessed, read, created, updated, or deleted by other applications within the same instance. These settings manage security between application scopes to prevent unauthorized cross-scope interactions.

❖Assign table access to application

- While creating a table it automatically creates a application and module for that table.
- Go to application navigator search for search project table application.
- Click on edit module.
- Give project member roles to that application.
- Search for task table2 and click on edit application.
- Give the project member and team member role for task table 2 application.

The top screenshot shows the 'Application Menu - project table' configuration page. It includes fields for Title (project table), Application (Global), Active (checked), Roles (Project Member), Category (Custom Applications), and Hint/Description text boxes. The bottom screenshot shows the 'Application Menu - Task Table 2' configuration page, with similar fields but Roles set to 'u_task_table_2_user, Project Member, Team Member'.

Step 8: Access control list

In ServiceNow, an ACL (Access Control List) is a security rule that governs who can access what data and perform which actions (Create, Read, Write, Delete) on tables, fields, or records, acting as granular security gates for the platform's data, ensuring users only see and modify what they are permitted to, often based on roles, conditions, and scripts.

- Open service now.
- Click on All >> search for ACL.
- Select Access Control (ACL) under system security.
- Click on elevate role.
- Click on new.
- Fill the following details to create a new ACL.
- Scroll down under requires role.
- Double click on insert a new row.

Give task table and team member role.

Click on submit.

The screenshot shows the ServiceNow 'Access Control' configuration page for the field 'u_task_table_2.u_status'. The 'Admin overrides' checkbox is checked. The 'Protection policy' is set to 'None'. The 'Name' is 'Task Table 2 [u_task_table_2]' and the 'Status' is 'Status'. The 'Description' field is empty. The 'Applies To' section shows 'No. of records matching the condition: 0' with buttons for 'Add Filter Condition' and 'Add OR Clause'. Below this is a field for 'choose field --' and a 'value --' field. The 'Conditions' section contains a blue box with text explaining decision types and a 'More Info' link. The 'Requires role' section shows a list of roles: 'Task Table' and 'Team Member', both with a red 'X' icon.

- Similarly create 4 acl for the following fields

The screenshot shows the ServiceNow 'Access Controls' list page. The table lists various access control rules with columns for Name, Decision Type, Operation, Type, Active, Updated by, and Updated. The table contains 13 rows of data.

Name	Decision Type	Operation	Type	Active	Updated by	Updated
u_project_table	Allow If	write	record	true	admin	2026-02-03 04:10:19
u_task_table_2	Allow If	read	record	true	admin	2026-02-03 05:25:01
u_task_table_2	Allow If	delete	record	true	admin	2026-02-03 05:25:01
u_task_table_2.u_assigned_to	Allow If	write	record	true	admin	2026-02-03 07:37:20
u_task_table_2	Allow If	write	record	true	admin	2026-02-03 05:25:01
u_task_table_2.u_task_name	Allow If	write	record	true	admin	2026-02-03 07:41:21
u_project_table	Allow If	delete	record	true	admin	2026-02-03 04:10:19
u_task_table_2.u_status	Allow If	write	record	true	admin	2026-02-03 07:17:04
u_task_table_2.u_task_id	Allow If	write	record	true	admin	2026-02-03 07:40:16
u_project_table	Allow If	create	record	true	admin	2026-02-03 04:10:19
u_task_table_2	Allow If	create	record	true	admin	2026-02-03 05:25:01
u_task_table_2.u_due_date	Allow If	write	record	true	admin	2026-02-03 07:38:49
u_project_table	Allow If	read	record	true	admin	2026-02-03 04:10:19

- Click on profile on top right side.
- Click on impersonate user.
- Select bob user.
- Go to all and select task table2 in the application menu bar.
- Comment and status fields are having the edit access.

ServiceNow

Task Table 2 - Create Created

Task Table 2
New record

Assigned To

Task ID

Due Date

Task Name

Status -- None --

Comments

Submit

Step 9: Create a Flow to Assign operations ticket to group

- Open service now.
- Click on All >> search for Flow Designer
- Click on Flow Designer under Process Automation.
- After opening Flow Designer Click on new and select Flow.
- Under Flow properties Give Flow Name as “task table”.
- Application should be Global.
- Click build flow.

Workflow Studio

New Flow

Let's get the details for your flow

Flow name * task table

Application * Global

Description

Describe your flow.

Show additional properties

Cancel Build flow

Next step:

Click on Add a trigger.

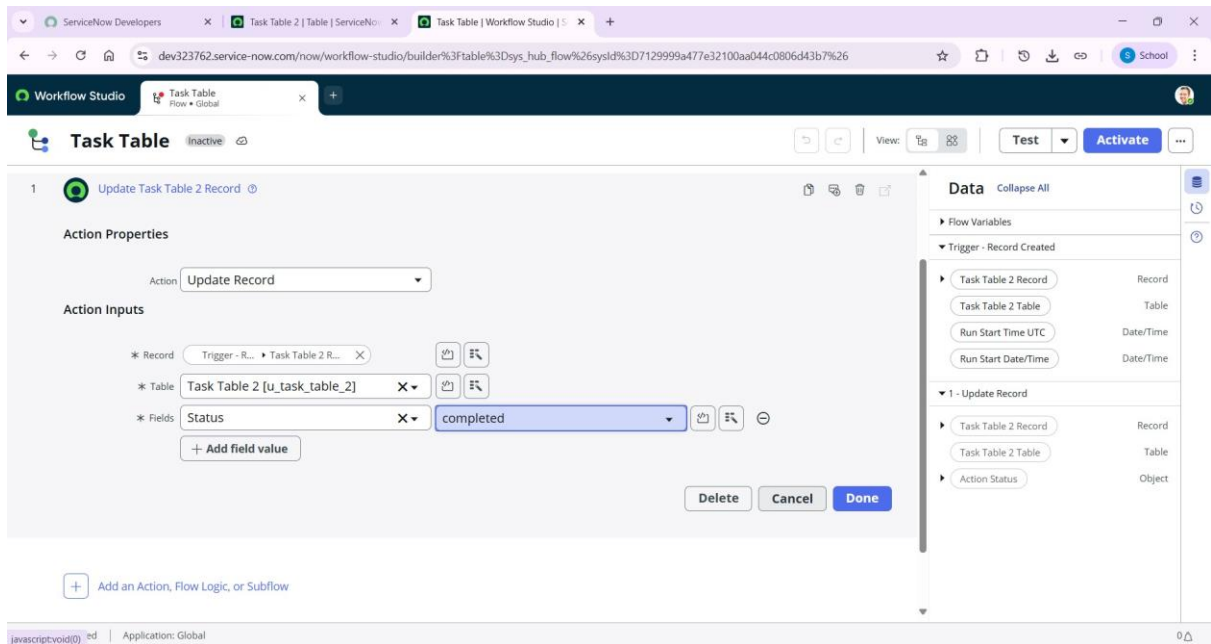
Select the trigger in that Search for “create record” and select that.

- Give the table name as “task table”.
- Give the Condition as:
 - ❖ Field: status: is: in progress
 - ❖ Field: comments: is: feedback
 - ❖ Field: assigned to: is: bob
- After that click on Done.

The screenshot displays the ServiceNow Workflow Studio interface for configuring a trigger. The trigger is named "Task Table 2 Created where (Status is in progress, and Comments is feedback, and Assigned To is bob)". The trigger type is "Created". The table selected is "Task Table 2 [u_task_table_2]". The condition is set to "All of these conditions must be met" and includes three criteria: "Status is in progress", "Comments is feedback", and "Assigned To is bob", all connected by "AND" logic. The right sidebar shows the "Data" navigation pane with various fields like "Task Table 2 Record", "Task Table 2 Table", "Run Start Time UTC", and "Run Start Date/Time". The bottom status bar indicates "Status: Modified" and "Application: Global".

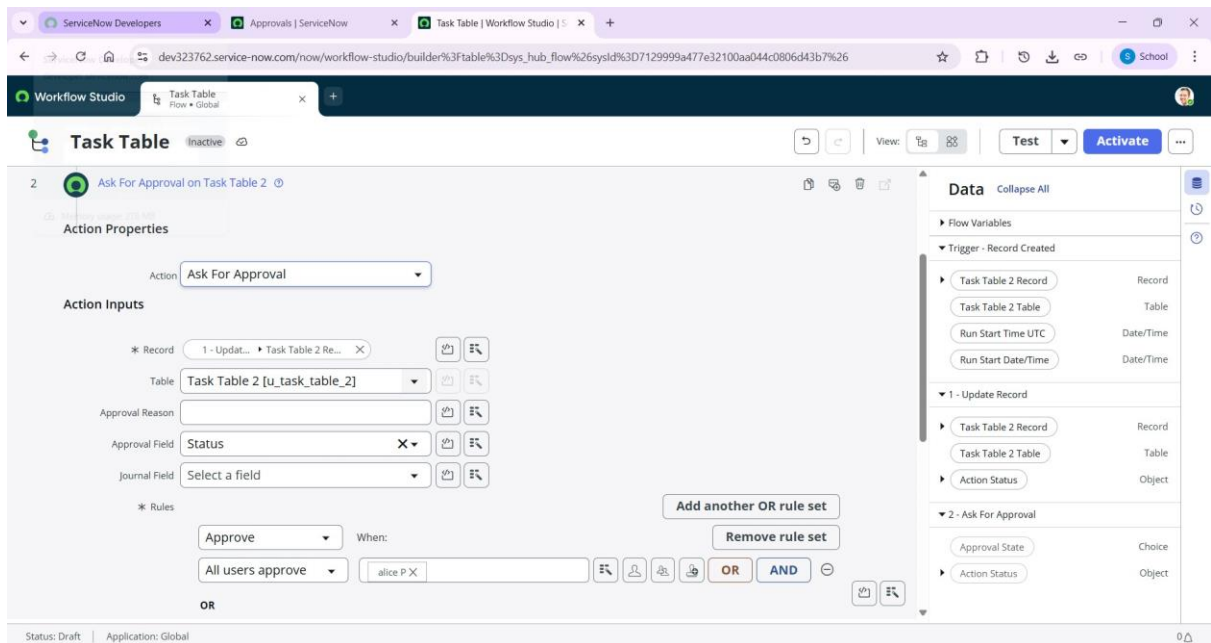
Next step:

- Click on Add an action.
- Select action in that, search for “update records”.
- In Record field drag the fields from the data navigation from Right Side(Data pill)
- Table will be auto assigned after that
- Add fields as “status” and value as “completed” • Click on Done.



Next step:

- Now under Actions.
- Click on Add an action.
- Select action in that, search for “ask for approval”.
- In Record field drag the fields from the data navigation from Right side
- Table will be auto assigned after that
- Give the approve field as “status”
- Give approver as alice p
- Click on Done.



Go to application navigator search for task table.

It status field is updated to completed.

ServiceNow Developers | Create Created | Task Table 2 | Task Table | Workflow Studio

dev323762.service-now.com/now/nav/ui/classic/params/target/u_task_table_2.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Du_task_table_2%26sys...

Task Table 2 - Create Created

Assigned To: bob

Task ID:

Due Date:

Task Name:

Status: completed

Comments:

Submit

- Go to application navigator and search for my approval.
- Click on my approval under the service desk.
- Alice p got approval request then right click on requested then select approved.

Approvals | ServiceNow | Task Table | Workflow Studio

dev323762.service-now.com/now/nav/ui/classic/params/target/sysapproval_approver_list.do%3Fsysparm_first_row%3D1%26sysparm_query%3D%26sys...

Approvals

State: Approver: Search

Actions on selected rows...

	State	Approver	Comments	Approval for	Created
☐	Search	Search	Search	Search	Search
☒	Approved	alice P		(empty)	2026-02-17 01:18:07
☐	Requested	Bernard Laboy		CHG0000058	2024-08-30 06:35:13
☐	Requested	Bernard Laboy		CHG0000007	2024-08-27 10:52:40
☐	Requested	Bernard Laboy		CHG0000089	2024-08-30 06:33:43
☐	Requested	Bernard Laboy		CHG0000076	2024-08-29 07:58:17
☐	Requested	Bernard Laboy		CHG0000096	2024-08-29 08:04:38
☐	Requested	Bernard Laboy		CHG0000043	2024-08-29 08:02:56
☒	Requested	Bernard Laboy		CHG0000087	2024-08-29 08:03:25
☐	Requested	Bernard Laboy		CHG0000042	2024-08-29 07:58:59
☐	Requested	Bernard Laboy		CHG0000084	2024-08-29 08:04:47
☐	Requested	Bernard Laboy		CHG0000037	2024-08-29 07:55:02
☐	Requested	Bernard Laboy		CHG0000057	2024-08-29 08:03:06
☐	Requested	Bernard Laboy		CHG0000093	2024-08-29 07:59:49
☐	Requested	Bernard Laboy		CHG0000092	2024-08-29 08:03:30
☐	Requested	Bernard Laboy		CHG0000083	2024-08-29 07:57:06
☐	Requested	Bernard Laboy		CHG0000065	2024-08-29 07:59:25

1 to 20 of 740

Step 10: Conclusion

This scenario highlights a structured approach to project management, showcasing the roles of Alice and Bob within a defined workflow. With Alice's oversight and Bob's execution, the team effectively collaborates to ensure project success. The use of tables organizes key information,

facilitating easy tracking of projects, tasks, and progress updates. Overall, this system promotes accountability, enhances communication, and leads to the successful completion of projects.